

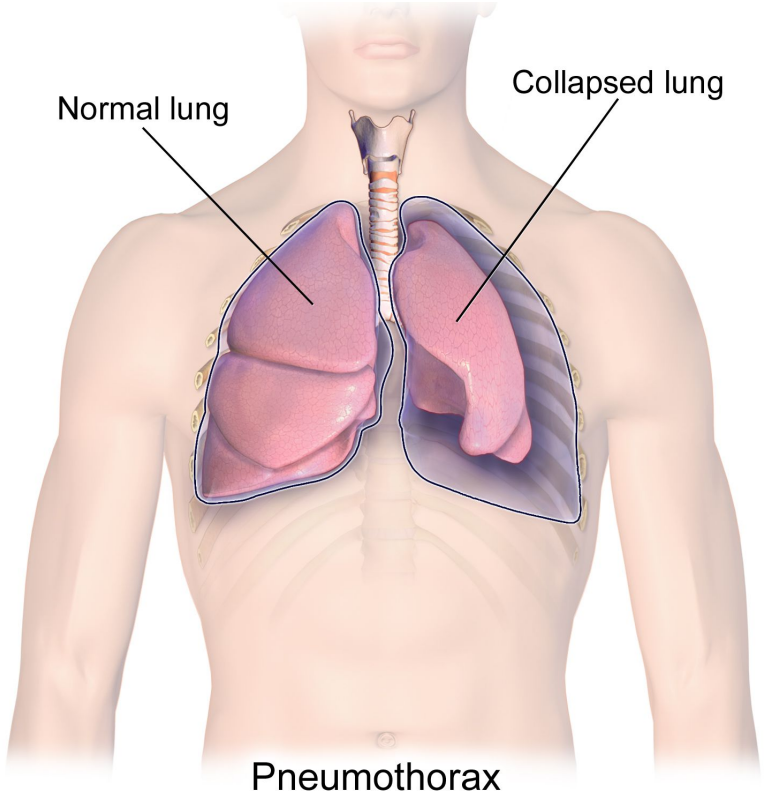


Dheemant Jallepalli, Jared Scott, Victor Chiu, Tushar Nayak, Pranav Kini, Defne Kudug

Self-Healing Seal for Chest Tube Removal

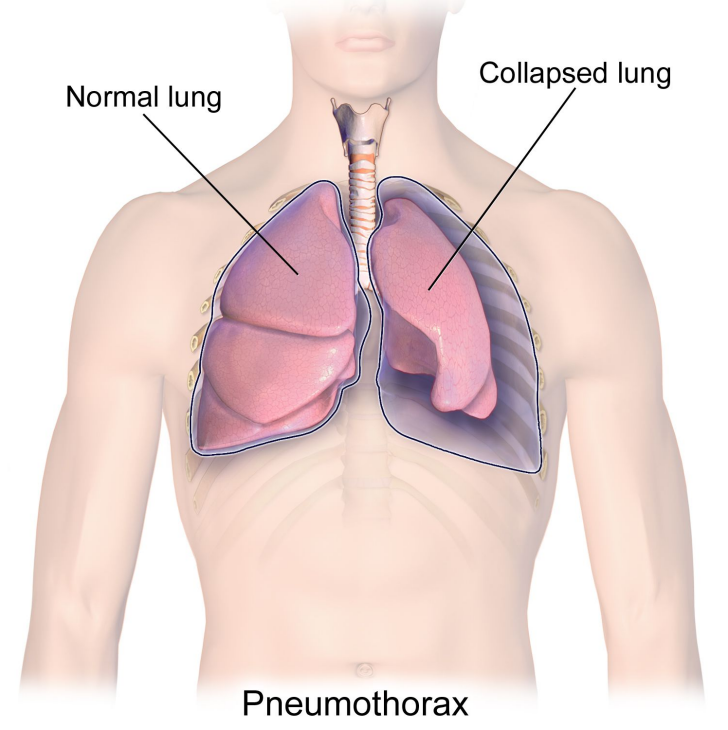
Background: Pneumothorax

- Air enters the pleural space, causing lung to collapse partially/fully
- Results when lung is damaged due to disease or injury
- Can reoccur during/after surgery once chest tube is removed



Pneumothorax

Background: Pneumothorax



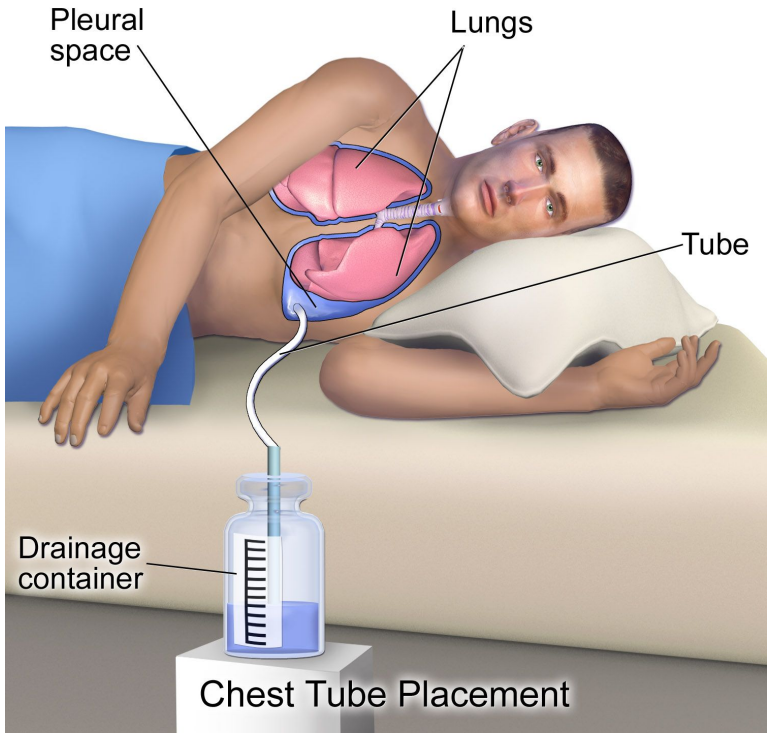
Air enters pleural space between lung/chest wall, causing lung to collapse

Typically occurs when lung is damaged due to disease or injury

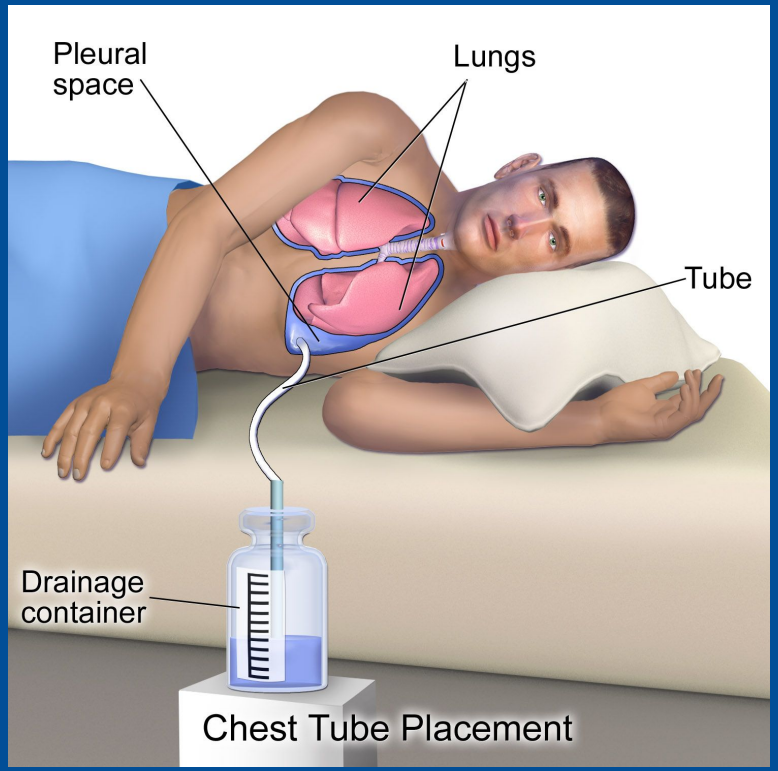
Can occur during/after lung surgery when chest tube is removed

Background: Current Surgical Practice

- “**Postoperative air leaks** are one of the most frequent complications after lung resection surgery” [1]
- A study of **497** patients undergoing chest tube removal after thoracic surgery reported that **81** developed pneumothorax as a result [1]
- Removal of chest tubes at the **end of inspiration with suction** and after **24-48 hours** is associated with a lower rate of recurrence of pneumothorax and a significantly shorter duration of hospital stay. [2]



Background: Current Surgical Practice



"**Postoperative air leaks** are one of the most frequent complications after lung resection surgery" [1]

A study of **497** patients undergoing chest tube removal after thoracic surgery reported that **81** developed pneumothorax as a result [1]

Proper timing and use of suction during tube removal correlate with a lower rates of recurrent pneumothorax

Challenge: Surgical Needs

Surgical Problem: Challenging removal of the chest tube post-operation causes increased risk of pneumothorax recurrence, patient discomfort, prolonged hospital stays, and additional clinical complications [3, 4, 5, 6].

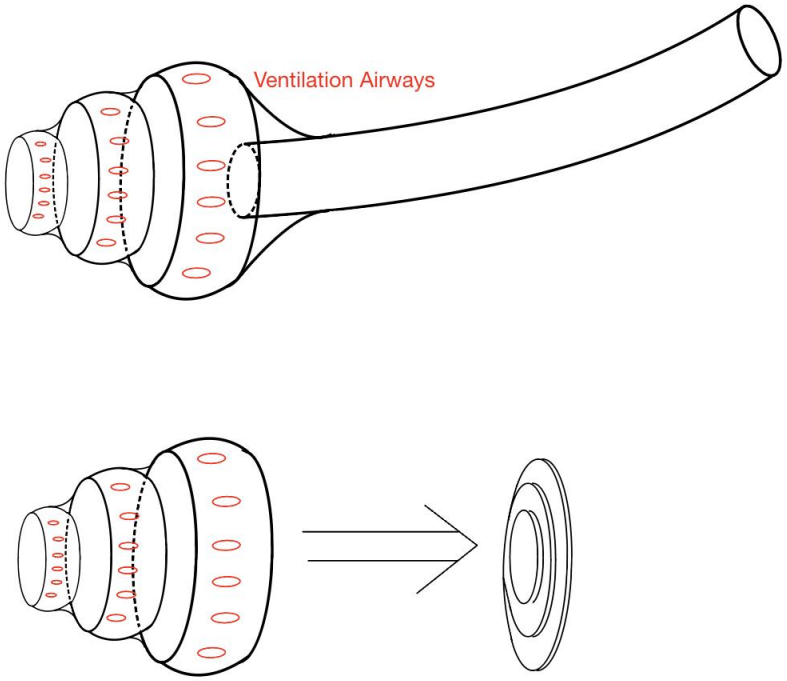
Surgical Characteristics:

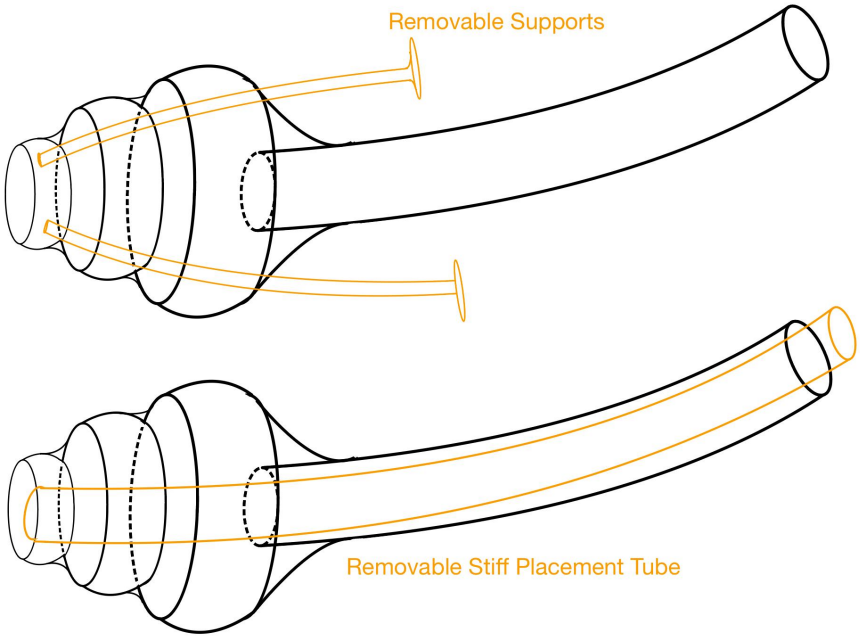
1. **Compatibility:** Compatible with standard chest tube (thoracostomy tubes) sizes (20-40 French) [7].
2. **Insertion:** Should not significantly alter current insertion techniques.
3. **Biocompatibility:** Materials must be biocompatible and non-irritating to patient tissues (ISO 10993 compliance) [8].
4. **Sterility:** Must maintain sterility throughout insertion and removal processes (ISO 11135 standards) [9].
5. **Radiopacity:** Incorporates radiopaque markers for X-ray visibility [10].
6. **Drainage:** Does not impede fluid or air drainage when in place.
7. **Fixation:** Secure fixation method to prevent accidental dislodgement.

Challenge: Post-Operational Needs

Usage Characteristics:

1. **Self-Sealing Mechanism:** Automatically seals the insertion site upon chest tube removal to prevent air leaks [11].
2. **Activation:** Sealing mechanism activates immediately upon tube withdrawal without additional clinician intervention.
3. **Removal Force:** Does not significantly increase required removal force.
4. **Seal Integrity:** Airtight seal maintained for 24-48 hours post-removal or until natural healing occurs [12].
5. **Visualization:** Enables clear visualization of drainage for patient monitoring.
6. **Compatibility with Suction:** Functional with standard chest drainage systems, including active suction systems.
7. **Ease of Use:** Intuitive operation requiring minimal additional healthcare professional training.
8. **Size Range:** Available in various sizes to fit different patient anatomies and clinical needs.
9. **Disposal:** Easily and safely disposable per standard medical waste protocols by CDC and EPA [13, 14].
10. **Storage:** Stable under standard hospital storage conditions, with a minimum shelf life of 2 years [15].





Solution: Internal Self-Healing Chest Tube Attachment



Thank you!

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